

problema no. 2.

$$q_0 = \varepsilon q_1 \mid \varepsilon q_3 = q_1 \mid q_3 = a_1 + a_3$$

$$q_1 = 0 q_0$$

$$q_2 = \varepsilon q_5 = a_5$$

$$q_3 = \varepsilon q_4 = q_4 \quad /$$

$$q_4 = \varepsilon q_5 = q_5 \quad /$$

$$q_5 = \varepsilon q_6 \mid \varepsilon q_{12} = q_6 \mid q_{12} = q_6 + a_{12} \quad /$$

$$q_6 = \varepsilon q_7 \mid \varepsilon q_9 = q_7 \mid q_9 = q_7 + a_9 \quad /$$

$$q_7 = 1 q_0 \quad /$$

$$q_8 = \varepsilon q_{11} = q_{11} \quad /$$

$$q_9 = \varepsilon q_{10} = q_{10} \quad /$$

$$q_{10} = \varepsilon q_{11} = q_{11} \quad /$$

$$q_{11} = \varepsilon q_{14} = q_{14} \quad /$$

$$q_{12} = \varepsilon q_{13} = q_{13} \quad /$$

$$q_{13} = \varepsilon q_{14} = q_{14} \quad /$$

$$q_{14} = \varepsilon q_{15} \mid \varepsilon q_{17} = q_{15} \mid q_{17} = q_{15} + q_{17} \quad /$$

$$q_{15} = 0 q_{14} \quad /$$

$$q_{16} = \varepsilon q_{15} \mid \varepsilon q_{17} = q_{15} \mid q_{17} = q_{15} + q_{17} \quad /$$

$$q_{17} = \varepsilon$$

$$q_{17} \rightarrow q_{14} = q_{15} + \varepsilon$$

$$q_{15} \rightarrow q_{16} = 0 q_{16} + \varepsilon = 0^*$$

$$q_{16} \rightarrow q_{15} = 0 \cdot 0^* = 0^+$$

$$q_{17} \rightarrow q_{14} = q_{15} + \varepsilon$$

$$q_{15} \rightarrow q_{14} = 0^+$$

$$q_{14} \rightarrow q_{13} = 0^+$$

$$q_{13} \rightarrow q_{12} = 0^+$$

$$q_{12} \rightarrow q_{11} = 0^+$$

$$q_{11} \rightarrow q_{10} = 0^+$$

$$q_{10} \rightarrow q_9 = 0^+$$

$$q_9 \rightarrow q_8 = 0^+$$

$$q_8 \rightarrow q_7 = 1 0^+$$

$$q_a \rightarrow q_u = q_z + 0^+$$

$$q_z \rightarrow q_u = 10^+ + 0^+$$

$$q_u \rightarrow q_c = q_u + 0^+$$

$$q_u \rightarrow q_c = 10^+ + 0^+, \quad 0^+ + 0^+ = 0^+$$

$$q_c \rightarrow q_u = 10^+ + 0^+$$

$$q_u \rightarrow q_z = 10^+ + 0^+$$

$$q_c \rightarrow q_z = 10^+ + 0^+$$

$$q_z \rightarrow q_1 = 0(10^+ + 0^+) = 010^+ + 00^+$$

$$q_z \rightarrow q_0 = q_1 + 10^+ + 0^+$$

$$q_1 \rightarrow q_0 = 010^+ + 00^+ + 10^+ + 0^+$$

$$\Rightarrow 010^+ + 00^+ + 10^+ + 0^+ = 010^+ + 10^+ + 0^+$$

$$\Rightarrow (010^+ | 10^+ | 0^+) //$$